

Radial ($\ell = 0$) and dipole ($\ell = 1$) oscillations of a static thin shell

Does the shell oscillate or collapse?

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Abstract

A thin spherical shell of perfect fluid, with a flat interior and a Schwarzschild exterior of mass M , can sit at rest at any radius $R > 2M$ if its surface pressure is $p = \sigma(1 - \sqrt{f})/4\sqrt{f}$, $f = 1 - 2M/R$. Its non-radial ($\ell \geq 2$) oscillations are known to contain a growing mode. Here we answer the complementary question: what happens to the lowest multipoles, the radial ($\ell = 0$) and dipole ($\ell = 1$) oscillations of the shell itself? They do not radiate, so they are true normal modes with real ω^2 , and their sign decides between oscillation and runaway. With $x \equiv \sqrt{f}$ and the adiabatic index Γ of the surface fluid we find, in closed form,

$$\frac{\omega_0^2 R^3}{M} = \frac{2f}{1 + \sqrt{f}} (\Gamma - \Gamma_1), \quad \Gamma_1 = \frac{1 + 2\sqrt{f} + 3f}{4f}; \quad \frac{\omega_1^2 R^3}{M} = \frac{6f}{1 + 3\sqrt{f}} (\Gamma - \Gamma_{\text{dip}}), \quad \Gamma_{\text{dip}} = \Gamma_1 - \frac{1}{6f},$$

plus a zero-frequency dipole mode (rigid translation). The odd dipole has no oscillation at all: it is a slow rigid rotation that drags the interior. For the equation of state used for this shell, $\Gamma = 2$, the shell is radially unstable for $R < 3.82M$ and dipole unstable for $R < 2.77M$; below $R = 2.37M$ no causal equation of state can make it radially stable. Following the unstable radial mode nonlinearly shows the shell collapsing to a black hole, the thin-shell version of the classic gravitational collapse problem. Units $G = c = 1$, time dependence $e^{-i\omega t}$, ω measured in exterior (Schwarzschild) time.

1 The question and why $\ell = 0, 1$ are special

The shell sits at areal radius R . Inside, $ds^2 = -dT^2 + dr^2 + r^2 d\Omega^2$; outside, $ds^2 = -f dt^2 + f^{-1} dr^2 + r^2 d\Omega^2$. The shell carries the surface stress tensor $S_{ab} = (\sigma + p)u_a u_b + p h_{ab}$ of a two-dimensional perfect fluid. The Israel junction conditions [1, 2]

$$[h_{ab}] = 0, \quad [K_{ab}] - h_{ab}[K] = -8\pi S_{ab} \quad (1)$$

fix the static configuration (Poisson [2], Eqs. 3.80–3.81; Ipser & Sikivie [3], Eqs. 3.7–3.9):

$$\sigma = \frac{1 - \sqrt{f}}{4\pi R}, \quad p = \kappa\sigma, \quad \kappa = \frac{1 - \sqrt{f}}{4\sqrt{f}}. \quad (2)$$

A perturbation of the fluid needs an equation of state. We use the one of Pitre, Schneider & Poisson [5]: $\Gamma = \partial \ln p / \partial \ln \mu$ at fixed entropy, with μ the rest-mass surface density. The first law $d\sigma = (\sigma + p) d\mu / \mu$ then gives the sound speed

$$v_s^2 = \frac{dp}{d\sigma} = \frac{\Gamma p}{\sigma + p} = \frac{\Gamma \kappa}{1 + \kappa}, \quad \text{causality } (v_s \leq 1): \quad \Gamma \leq \Gamma_{\text{caus}} = \frac{1 + 3\sqrt{f}}{1 - \sqrt{f}}. \quad (3)$$

What is different at $\ell = 0, 1$. For $\ell \geq 2$ the shell's oscillations couple to gravitational waves, and the problem is a scattering or quasinormal-mode problem with complex frequencies. That problem has four matter modes per ℓ , one of them growing [5]. For $\ell = 0$ and $\ell = 1$ there is no radiation:

- $\ell = 0$: by Birkhoff's theorem the exterior stays Schwarzschild. The only non-gauge vacuum perturbation is a change δM of the mass, which is static.
- $\ell = 1$, even: every vacuum perturbation is pure gauge, both in flat space and in Schwarzschild (Martel & Poisson [4], Sec. IV D). Physically this is momentum conservation: the centre of mass cannot oscillate.
- $\ell = 1$, odd: the only non-gauge vacuum perturbation is a change δJ of the angular momentum (linearized Kerr), which is static ([4], Sec. V E).

The whole dynamics is therefore in the shell's matter and position, the frequencies are real or purely imaginary, and ω^2 is real: $\omega^2 > 0$ is an oscillation, $\omega^2 < 0$ an exponential runaway.

2 Radial oscillations, $\ell = 0$: exact equation of motion

Result. The radial problem can be solved exactly: the spherical shell moves in a one-dimensional effective potential.

For a spherical shell $r = R(\tau)$, with τ its proper time, Eq. (1) gives $\sigma = [\sqrt{1 + \dot{R}^2} - \sqrt{f(R) + \dot{R}^2}]/4\pi R$ [3]. The conservation law $D_b S^{ab} = 0$ is the first law $d(\sigma A) + p dA = 0$ with $A = 4\pi R^2$, so $d\sigma/dR = -2(\sigma + p)/R$. With $m(R) \equiv 4\pi R^2 \sigma(R)$, write $\alpha = \sqrt{1 + \dot{R}^2}$ and $\beta = \sqrt{f + \dot{R}^2}$. Then $\alpha - \beta = m/R$ and $\alpha^2 - \beta^2 = 2M/R$, so $\alpha = M/m + m/2R$ and

$$\dot{R}^2 + V(R) = 0, \quad V(R) = 1 - \left(\frac{M}{m} + \frac{m}{2R} \right)^2. \quad (4)$$

This is exact (nonlinear). A static shell sits where $V = V' = 0$. Small displacements obey $\delta \ddot{R} = -\frac{1}{2} V'' \delta R$. Using $m' = -8\pi R p$ and $m'' = -8\pi p + 16\pi v_s^2(\sigma + p)$, and converting to exterior time ($d\tau = \sqrt{f} dt$ for a static shell), we get

$$\frac{\omega_0^2 R^3}{M} = \frac{4\Gamma f - 3f - 2\sqrt{f} - 1}{2(1 + \sqrt{f})} = \frac{2f}{1 + \sqrt{f}} (\Gamma - \Gamma_1), \quad \Gamma_1 = \frac{1 + 2\sqrt{f} + 3f}{4f} = \frac{3}{2} + 4\kappa + 4\kappa^2 \quad (5)$$

(code/10_potential.py). The shell is radially stable if and only if $\Gamma > \Gamma_1(R)$.

- The marginal index Γ_1 coincides *identically* with the turning point of the equilibrium sequence, $dM/d\mu = 0$, of [5] (their Eq. 3.16) and LeMaitre & Poisson [6]. Those works gave the stability criterion. Equation (5) gives the actual frequency and growth rate.
- Newtonian limit ($C = M/R \rightarrow 0$): $\omega_0^2 R^3/M = \Gamma - \frac{3}{2} + (\frac{5}{4} - \frac{3}{2}\Gamma)C + O(C^2)$. The leading term is the Newtonian result $\Gamma - 3/2$ [5]. General relativity raises the required stiffness, exactly as for stars.
- **For $\Gamma = 2$ the shell is radially unstable for $R < 3.8165M$ ($C > 0.2620$).** This includes the compact shells $R = 2.2M$, $2.5M$ and $3M$ used for the trapped modes and echoes of this system.
- Γ_1 exceeds the causal bound Γ_{caus} of Eq. (3) for $R < 2.370M$. There, *no causal equation of state can make the shell radially stable*. At the energy-condition limit $R = 25M/12$, $\Gamma_1 = 9.5$.

3 Dipole oscillations, $\ell = 1$

3.1 Even parity: derivation from the Israel equations

Result. Because even $\ell = 1$ vacuum perturbations are pure gauge, the metric can be left unperturbed on both sides. All the physics is in how the shell is embedded in each side and in how its fluid moves.

On each side we use coordinates in which the metric is exactly the background one. The gauge transformations of the two sides are independent, and they are absorbed in the two embeddings. With intrinsic coordinates $y^a = (\tau, \theta, \phi)$, $Y = \cos \theta$, and all amplitudes $\propto e^{-i\Omega\tau}$,

$$t_{\pm} = \frac{\tau}{\sqrt{F_{\pm}}} + \epsilon A_{\pm} Y, \quad r_{\pm} = R + \epsilon X_{\pm} Y, \quad \theta_{\pm} = \theta + \epsilon B_{\pm} \partial_{\theta} Y, \quad (6)$$

with $F_- = 1$ inside and $F_+ = f$ outside. The fluid has $\sigma \rightarrow \sigma + \epsilon s Y$, $p \rightarrow p + \epsilon v_s^2 s Y$ and a tangential velocity $u^{\theta} = \epsilon w \partial_{\theta} Y$. From Eq. (6) we compute, on each side, the induced metric and the extrinsic curvature $K_{ab} = -n_{\mu}(\partial_b e_a^{\mu} + \Gamma_{\nu\rho}^{\mu} e_a^{\nu} e_b^{\rho})$ to first order. Then we impose Eq. (1): continuity of h_{ab} and the Israel equations, for the $\tau\tau$, $\tau\theta$, $\theta\theta$ and $\phi\phi$ components (code/israel_low_1.py, SymPy). A reparametrization of the shell $y^a \rightarrow y^a + \epsilon \zeta^a$ shifts $A_{\pm}$ and $B_{\pm}$ together. We fix it with $A_- = B_- = 0$ and have checked that fixing it on the exterior side instead gives identical frequencies. This leaves six unknowns, $(X_-, A_+, X_+, B_+, s, w)$, and six independent equations. The zeroth order of every equation vanishes for the background (2).

The same code with $Y = 1$ solves the $\ell = 0$ problem. It reproduces Eq. (5) identically, an independent check of the whole construction (normal, curvature, fluid terms). For $\ell = 1$ the determinant is

$$\det \propto \Omega^2 \left[4\Omega^2 x^2 (3x + 1) + 12\Gamma x^2 (x^2 - 1) - 9x^4 - 6x^3 + 8x^2 + 6x + 1 \right], \quad x = \sqrt{f}, \quad (7)$$

which gives, with $\omega = \sqrt{f} \Omega$,

$$\frac{\omega_1^2 R^3}{M} = \frac{12\Gamma f - 9f - 6\sqrt{f} - 1}{2(1 + 3\sqrt{f})} = \frac{6f}{1 + 3\sqrt{f}} (\Gamma - \Gamma_{\text{dip}}), \quad \Gamma_{\text{dip}} = \frac{(1 + 3\sqrt{f})^2}{12f} = \Gamma_1 - \frac{1}{6f} \quad (8)$$

and a double root $\Omega = 0$.

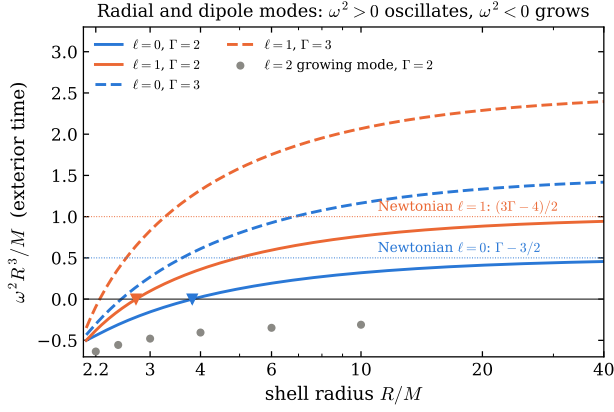


Figure 1: $\omega^2 R^3/M$ of the radial (blue) and dipole (orange) modes for $\Gamma = 2$ (solid) and $\Gamma = 3$ (dashed). Triangles: marginal radii for $\Gamma = 2$. Grey: the growing $\ell = 2$ matter mode, $-(\text{Im } \omega)^2 R^3/M$. Dotted: Newtonian limits.

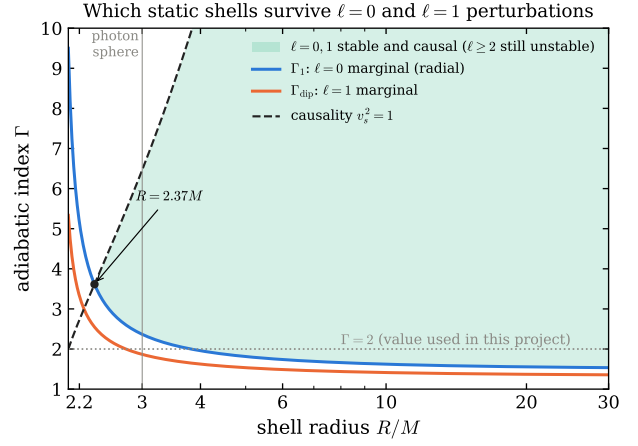


Figure 2: Stability in the (R, Γ) plane. Above Γ_1 : radially stable; above Γ_{dip} : dipole stable; below the dashed line: causal. Shaded: stable against $\ell = 0, 1$ with a causal equation of state.

- **Zero mode.** Its null vector is $X_- = -RB_+$, with $X_+ = s = w = 0$: the shell is *rigidly translated* relative to the interior flat space and stays centred in the exterior. The double root corresponds to the translation and the uniform motion (boost). This is the relativistic version of the zero mode found by [5] (their Eq. 10.58).
- **Oscillating mode.** The shell's geometric centre moves while its surface density develops a dipole (Fig. 4). The centre of mass stays fixed: in the Newtonian limit the eigenvector satisfies $X_+ + sR/3\sigma = 0$, with a residual that vanishes as M/R . The restoring force is the pressure of the compressed side.
- **Newtonian limit:** $\omega_1^2 R^3/M \rightarrow \frac{1}{2}(3\Gamma - 4)$, exactly Eq. 10.57 of [5]. It also matches our own Newtonian shell equations evaluated at $\ell = 1$ (`code/check_newtonian.py`). Relativistic correction: $+(\frac{3}{2} - \frac{15}{8}\Gamma)C$.
- Since $\Gamma_{\text{dip}} < \Gamma_1$ at every radius, **a radially stable shell is always dipole stable**. For $\Gamma = 2$ the dipole mode is unstable only for $R < 2.767M$. A causal equation of state can stabilize it down to $R = 9M/4$ exactly.

3.2 Odd parity: no oscillation, only rotation

For odd parity the conservation law $D_b S^{ab} = 0$ gives $\partial_\tau u_A = 0$ at every ℓ . The shell's matter cannot oscillate axially, so there is *no odd matter mode at any ℓ* , and no odd dipole oscillation. The only odd $\ell = 1$ solution is stationary: a slow rigid rotation. Outside, $g_{t\phi} = -(2J/r)\sin^2\theta$. Inside, the flat space rotates rigidly. Continuity of $h_{\tau\phi}$ and the $\tau\phi$ Israel equation give (`code/11_odd_rotation.py`)

$$\Omega_{\text{drag}} = \frac{2J}{R^3}, \quad \frac{\Omega_{\text{drag}}}{\Omega_{\text{shell}}} = \frac{(1 - \sqrt{f})(1 + 3\sqrt{f})}{1 + 2\sqrt{f}} \xrightarrow{M/R \rightarrow 0} \frac{4M}{3R}, \quad \xrightarrow{R \rightarrow 2M} 1. \quad (9)$$

Here Ω_{drag} is the angular velocity at which the interior's local inertial frames rotate and Ω_{shell} that of the fluid, both in exterior time. The weak-field limit is Thirring's classic frame-dragging result [7]. The limit $R \rightarrow 2M$ is complete dragging, as found by Brill & Cohen [8]. The same equation returns $J = \frac{8\pi}{3}R^2(\sigma + p)U$, the shell's own angular momentum.

4 Beyond linear order: the unstable shell collapses

Result. The exact potential (4) lets us follow the radial instability to the end. The unstable shell either collapses through its horizon or makes a large bounded excursion. This is the thin-shell analogue of the collapse of a spherical body [10].

To go nonlinear we need the full equation of state. We take a polytrope [5], $p = K\mu^\Gamma$ and $\sigma = \mu + p/(\Gamma - 1)$, with rest-mass conservation $\mu \propto R^{-2}$, and choose K so that the shell of Eq. (2) at R_0 is an equilibrium. The shell is released from rest at $R_0(1 \pm \delta)$ and $\ddot{R} = -V'/2$ is integrated (`code/10_nonlinear.py`; Figs. 5, 6).

- $R_0 = 6M$ (stable). The shell oscillates about R_0 . The measured frequency, $\Omega = 0.036579/M$ in proper time for a 1% amplitude, agrees with Eq. (5) ($0.036588/M$) to 0.03%. V has a barrier at $R \approx 2.97M$, so a large enough inward kick collapses even this shell.

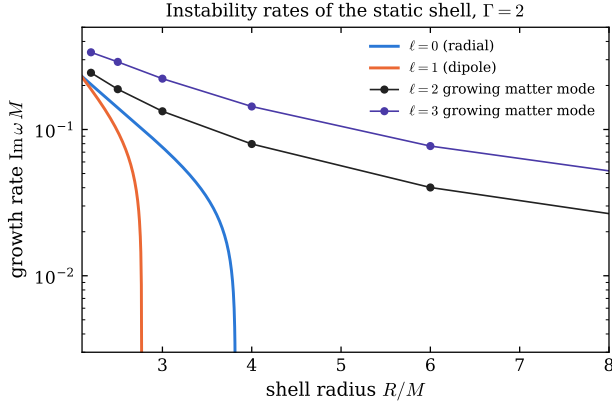


Figure 3: Growth rates for $\Gamma = 2$: radial and dipole (this work) and the growing $\ell = 2, 3$ matter modes of the same shell.

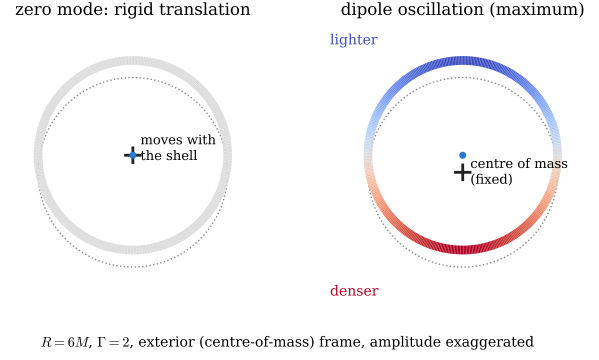


Figure 4: The two even dipole modes. Left: zero mode (translation). Right: dipole oscillation, colour = surface density; the centre of mass (cross) does not move.

Table 1: Marginal indices and frequencies. Γ_{caus} is the largest causal Γ . ωM is real for an oscillation and imaginary for a growing mode. Last column: the growing $\ell = 2$ matter mode of the same shell, from the full even-parity junction with radiation.

R/M	marginal / causal Γ			$\ell = 0, \Gamma = 2$		$\ell = 1, \Gamma = 2$		$\ell = 2, \Gamma = 2$
	Γ_1	Γ_{dip}	Γ_{caus}	$\omega^2 R^3/M$	ωM	$\omega^2 R^3/M$	ωM	ωM (growing)
2.2	5.158	3.325	2.73	-0.4412	0.2036 i	-0.3795	0.1888 i	0.2443 i
2.5	3.118	2.285	4.24	-0.3090	0.1406 i	-0.1459	0.0966 i	0.1886 i
3	2.366	1.866	6.46	-0.1547	0.0757 i	+0.0981	0.0603	0.1332 i
4	1.957	1.624	10.66	+0.0251	0.0198	+0.3616	0.0752	0.0796 i
6	1.737	1.487	18.80	+0.1928	0.0299	+0.5944	0.0525	0.0401 i
10	1.622	1.413	34.89	+0.3197	0.0179	+0.7647	0.0277	0.0176 i
100	1.510	1.340	394.99	+0.4824	0.0007	+0.9774	0.0010	—

- $R_0 = 3M$, pushed inward by 10^{-3} . The shell leaves its equilibrium exponentially (linear e-folding time $7.6M$) and crosses $r = 2M$, forming a black hole, after $\tau = 45.6M$ of proper time. A distant observer sees it freeze at the horizon.
- $R_0 = 3M$, pushed outward. The shell does not escape. It is gravitationally bound: its rest mass exceeds M by 3.6%. It expands to $9.5M$, falls back, and lingers near the unstable equilibrium. Any further inward perturbation there sends it into the black hole.

5 How important are these modes compared with $\ell \geq 2$?

Figure 3 and Table 1 compare the growth rates for $\Gamma = 2$.

- Wherever the radial mode is unstable, it grows somewhat more slowly than the $\ell = 2$ matter mode: by a factor 0.83 at $2.2M$ and 0.57 at $3M$. The time scale is the same, $(R^3/M)^{1/2}$, and is much longer than the wave periods $\sim R$. The adiabatic picture of scattering off a static shell is therefore not changed.
- The radial instability matters because of its outcome. An $\ell \geq 2$ runaway deforms the shell. An $\ell = 0$ runaway is *total collapse to a black hole*, or a large pulsation.
- The $\ell \geq 2$ instability is present for every Γ [5]. The $\ell = 0, 1$ instabilities, by contrast, disappear for a stiff enough causal fluid, but only if $R > 2.37M$. The ultracompact shells that produce echoes ($R < 3M$) therefore need Γ between Γ_1 (2.4 to 3.6) and the causal limit to avoid collapse.

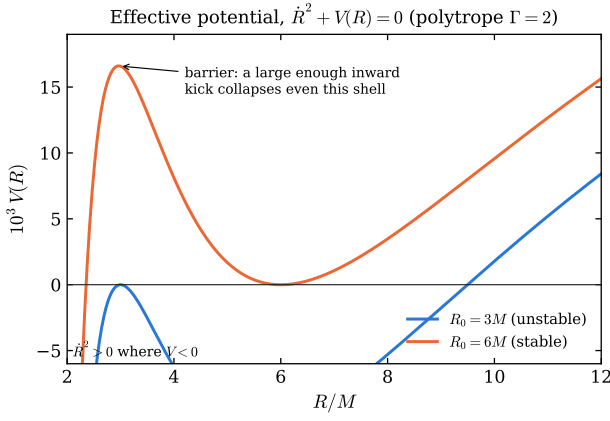


Figure 5: Exact effective potential (4) of a polytropic ($\Gamma = 2$) shell in equilibrium at $3M$ (a maximum: unstable) and $6M$ (a minimum: stable).

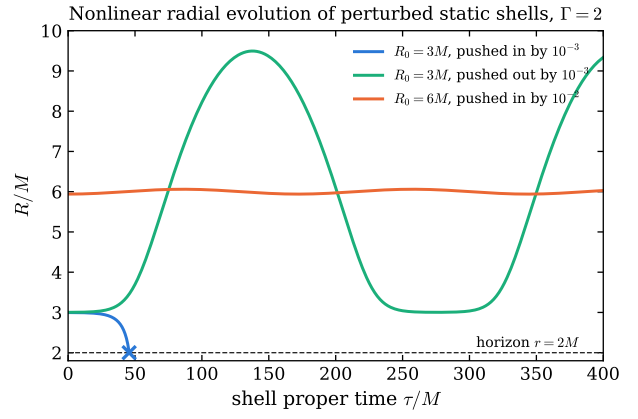


Figure 6: Nonlinear radial evolution. The unstable shell collapses through $r = 2M$ (cross) or makes a large bounded excursion; the stable one oscillates.

6 Validation

check	result
$\ell = 0$: exact potential vs linearized Israel equations (two independent derivations)	identical (symbolic difference 0)
$\ell = 0$ marginal index vs turning point $dM/d\mu = 0$ of [5, 6]	identical
Newtonian limits: $\ell = 0$, $\Gamma - \frac{3}{2}$; $\ell = 1$, $\frac{1}{2}(3\Gamma - 4)$ and 0	agree with [5] Sec. 10H and with our own Newtonian equations
background Israel equations at zeroth order (16 components)	all vanish
gauge fixing interior ($A_- = B_- = 0$) vs exterior ($A_+ = B_+ = 0$)	same frequencies
redundant components ($\phi\phi$, second angle)	add no rank (random numerical tests)
$\ell = 1$ zero mode	rigid translation (as [5] Eq. 10.58)
$\ell = 1$ oscillation: centre of mass fixed (Newtonian)	residual $\propto M/R$ (5×10^{-5} at $R = 10^4 M$)
nonlinear vs linear radial frequency ($R_0 = 6M$)	0.03%
odd $\ell = 1$: weak-field and $R \rightarrow 2M$ limits of the frame dragging	$4M/3R$ (Thirring), 1 (complete dragging)

7 Conclusions

1. The static fluid shell has one radial mode and, per m , one oscillating even dipole mode plus a translation zero mode. Both have closed-form frequencies, Eqs. (5) and (8). The odd dipole never oscillates; it is a slow rotation that drags the interior.
2. Radial stability requires $\Gamma > \Gamma_1 = (1 + 2\sqrt{f} + 3f)/4f$ and dipole stability $\Gamma > \Gamma_1 - 1/6f$, so the radial mode is always the more dangerous one. For the $\Gamma = 2$ shell the radial mode is unstable inside $3.82M$ and the dipole inside $2.77M$. Inside $2.37M$ no causal fluid can hold the shell up against radial collapse.
3. The unstable radial mode is the onset of gravitational collapse. Followed nonlinearly, the shell falls through its Schwarzschild radius in a few tens of M of proper time, or rebounds as a large bounded pulsation.

Next step. The natural continuation is to put the $\ell \geq 2$ gravitational perturbations on top of this collapsing background: first the odd parity, whose junction is a δ -barrier of strength $\sqrt{f}(1 - \sqrt{f})/R$ that can simply follow $R(\tau)$. This gives the waveform emitted while the shell collapses, and shows how its ringing turns into black-hole ringing as the horizon forms. It is the thin-shell analogue of the calculation of Cunningham, Price & Moncrief [9] for a collapsing dust ball.

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